

TØ 12 – Mikroøkonomi 2 – 2024

Problems for Tutorial

Problem 1

Consider a setting where a government is choosing how to design a tax system. In this setting the income, q , of worker i is determined as a function of his effort e and type θ as $q = \theta e$.

θ represents the productivity type of the individual. There are two types of individuals: a low type θ_L and a high type θ_H (where $\theta_H > \theta_L$). The probability that a random individual is of low type is β and he is of high type with probability $1 - \beta$.

All individuals in this economy have the same increasing and strictly concave von Neumann-Morgenstern utility function given by $u(q - t - \psi(e))$ where t is the net tax (or subsidy) which the individual has to pay (receive) and $\psi(e)$ is an increasing and convex cost function.

The government is utilitarian and wants to maximize the sum of utilities in the economy (or simply the expected utility of individual i without knowing his type). To achieve this goal the government can redistribute income across individuals through the tax scheme satisfying the following budget constraint: $\beta t_L + (1 - \beta)t_H = 0$ where t_L is the tax revenue (cost) from L-types and t_H is the tax revenue (cost) from the H-types.

Assume further that the government can simply require participation from its workers and therefore we do not need to consider participation constraints below.

First best: Assume that the government observes the type θ_i of individual i .

1. Formulate the government's maximization problem which determines the choice of t_L, e_L, t_H, e_H under the first best.
2. Characterize the first best solution and interpret the first order conditions.

1

Government maximises individual utility:

$$\max \beta u(\theta_L e_L - t_L - \psi(e_L)) + (1 - \beta)u(\theta_H e_H - t_H - \psi(e_H))$$

St

$$\beta t_L + (1 - \beta)t_H = 0 \quad (BC)$$

Budget constraint must be binding so solve for t_L :

$$\beta t_L + (1 - \beta)t_H = 0 \Leftrightarrow t_L = -\frac{1 - \beta}{\beta} t_H$$

Explanation: the low type pays negative tax (a subsidy) if the high type pays positive tax, i.e., if $t_H > 0$

Thus, the maximisation problem is now (with no constraint as we have included it in the max)

$$\max \beta u \left(\theta_L e_L + \frac{1-\beta}{\beta} t_H - \psi(e_L) \right) + (1-\beta) u (\theta_H e_H - t_H - \psi(e_H))$$

2

FOCs are

$$\begin{aligned} \frac{\partial}{\partial t_H} &= \beta \frac{1-\beta}{\beta} \cdot u' \left(\theta_L e_L + \frac{1-\beta}{\beta} t_H - \psi(e_L) \right) - (1-\beta) u' (\theta_H e_H - t_H - \psi(e_H)) = 0 \\ \Leftrightarrow u' \left(\theta_L e_L + \frac{1-\beta}{\beta} t_H - \psi(e_L) \right) &= u' (\theta_H e_H - t_H - \psi(e_H)) \end{aligned}$$

Since $\beta \frac{1-\beta}{\beta} = 1 - \beta$

Explanation: the different type's marginal utility is identical in order to maximise welfare

$$\begin{aligned} \frac{\partial}{\partial e_L} &= \beta u' \left(\theta_L e_L + \frac{1-\beta}{\beta} t_H - \psi(e_L) \right) (\theta_L - \psi'(e_L)) = 0 \Leftrightarrow \theta_L = \psi'(e_L) \\ \frac{\partial}{\partial e_H} &= (1-\beta) u' (\theta_H e_H - t_H - \psi(e_H)) (\theta_H - \psi'(e_H)) = 0 \Leftrightarrow \theta_H = \psi'(e_H) \end{aligned}$$

Since $\beta, 1 - \beta$ and $u'(\dots) > 0$ by definition

Notice that $\theta_H > \theta_L \Rightarrow \psi'(e_H) > \psi'(e_L) \Rightarrow \psi(e_H) > \psi(e_L) \Rightarrow e_H > e_L$

Second best A: Income contingent taxes: Assume now that the government can specify taxes dependent on the realization of income q but do not directly observe the individual components θ and e . This means that an individual with productivity θ_j can always pretend to be of type θ_i by effort level $\frac{\theta_i e_i}{\theta_j}$.

3. Formulate two IC constraints for the analysis under the second best, and explain why we can still cast the problem of the government as one of choosing levels of t_L, e_L, t_H, e_H .

Hint: Because the utility function has only one argument/composite index ($q - t - \psi(e)$) the constraints can simply be formulated as comparing the index (and not the utility) across alternatives.

4. Show that the first best contract (i.e. with the first best level of q_i and t_i) from above is not implementable under the second best.
5. Formulate the government's maximization problem under the second best.
6. Briefly explain the meaning and intuition of the term "no distortions at the top" (2-3 sentences). And show that the first order condition for e_H implies no "distortion at the top".

Second best B: Effort contingent taxes: Assume now that the government can directly specify the effort level required by individual i but do not directly observe the individual components θ_i and q .

7. Formulate the government's maximization problem under the second best.

Part 2 – Second best

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Now, second best, where $q_i = \theta_j \cdot \frac{\theta_i e_i}{\theta_j}, i \neq j$

$$\max \beta u(\theta_L e_L - t_L - \psi(e_L)) + (1 - \beta)u(\theta_H e_H - t_H - \psi(e_H))$$

$$\beta t_L + (1 - \beta)t_H = 0 \quad (BC)$$

$$\theta_L e_L - t_L - \psi(e_L) \geq \theta_L \cdot \frac{\theta_H e_H}{\theta_L} - t_H - \psi\left(\frac{\theta_H e_H}{\theta_L}\right) \quad (ICL)$$

$$\theta_H e_H - t_H - \psi(e_H) \geq \theta_H \cdot \frac{\theta_L e_L}{\theta_H} - t_L - \psi\left(\frac{\theta_L e_L}{\theta_H}\right) \quad (ICH)$$

We can still cast the problem of the government as one of choosing levels of taxes and effort since $q_L' = \theta_H \cdot \frac{\theta_L e_L}{\theta_H}$ and $q_H' = \theta_L \cdot \frac{\theta_H e_H}{\theta_L}$ as the taxes are defined as effort times type: $q = \theta e$ (just from definition in the assignment)

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In first best, we found that (remember that $q_i = \theta_i e_i, i \in \{L, H\}$):

$$u'(q_L^{FB} - t_L^{FB} - \psi(e_L^{FB})) = u'(q_H^{FB} - t_H^{FB} - \psi(e_H^{FB}))$$

Implying that

$$q_L^{FB} - t_L^{FB} - \psi(e_L^{FB}) = q_H^{FB} - t_H^{FB} - \psi(e_H^{FB})$$

But the good—or high—type will now pretend to be the bad—or low—type:

$$q_H^{FB} - t_H^{FB} - \psi(e_H^{FB}) = q_L^{FB} - t_L^{FB} - \psi(e_L^{FB}) < q_H^{FB} - t_H^{FB} - \psi\left(\frac{\theta_L e_L^{FB}}{\theta_H}\right)$$

Since $\theta_H > \theta_L \Leftrightarrow \frac{\theta_L}{\theta_H} < 1$ and hence, $\psi(e_L^{FB}) > \psi\left(\frac{\theta_L e_L^{FB}}{\theta_H}\right)$

So, to sum up: the FB contract is not implementable under SB

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This was answered in question 3

6

No distortion at the top means that the effort of the good type — sometimes also called high type — is not distorted under SB since it is too costly for the low type to pretend to be the high type.

I use $t_L = -\frac{1-\beta}{\beta}t_H$ to show that ICH binds:

$$\begin{aligned} \theta_H e_H - t_H - \psi(e_H) &\geq \theta_H \cdot \frac{\theta_L e_L}{\theta_H} - t_L - \psi\left(\frac{\theta_L e_L}{\theta_H}\right) \\ \Leftrightarrow \theta_H e_H - t_H - \psi(e_H) &\geq \theta_H \cdot \frac{\theta_L e_L}{\theta_H} + \frac{1-\beta}{\beta}t_H - \psi\left(\frac{\theta_L e_L}{\theta_H}\right) \\ \Leftrightarrow \theta_H e_H - \frac{1}{\beta}t_H - \psi(e_H) &\geq \theta_H \cdot \frac{\theta_L e_L}{\theta_H} - \psi\left(\frac{\theta_L e_L}{\theta_H}\right) \end{aligned}$$

Because if it did not bind, P could increase tax, t_H until the constraint would be binding

ICL does not bind since from the binding ICH:

$$\begin{aligned} \theta_H e_H - t_H - \psi(e_H) &= \theta_H \cdot \frac{\theta_L e_L}{\theta_H} - t_L - \psi\left(\frac{\theta_L e_L}{\theta_H}\right) \\ \Leftrightarrow \theta_H e_H - t_H - \psi(e_H) &= \theta_L e_L - t_L - \psi\left(\frac{\theta_L e_L}{\theta_H}\right) \\ \theta_H e_H - t_H - \psi\left(\frac{\theta_H}{\theta_L} \cdot e_H\right) &< \theta_L e_L - t_L - \psi\left(\frac{\theta_H}{\theta_L} \cdot \frac{\theta_L e_L}{\theta_H}\right) \\ \Leftrightarrow \theta_H e_H - t_H - \psi\left(\frac{\theta_H}{\theta_L} \cdot e_H\right) &< \theta_L e_L - t_L - \psi(e_L) \end{aligned}$$

Where I have used the fact that since $\theta_H > \theta_L \Leftrightarrow \frac{\theta_H}{\theta_L} > 1$

Which implies that $\psi\left(\frac{\theta_H}{\theta_L} \cdot e_H\right) > \psi(e_L)$ since $\psi(e_H) > \psi(e_L)$ and $\frac{\theta_H}{\theta_L} > 1$

ICH binds so

$$\theta_H e_H - t_H - \psi(e_H) = \theta_H \cdot \frac{\theta_L e_L}{\theta_H} - t_L - \psi\left(\frac{\theta_L e_L}{\theta_H}\right)$$

Rewrite

$$\begin{aligned} -t_H &= \theta_H \cdot \frac{\theta_L e_L}{\theta_H} - t_L - \psi\left(\frac{\theta_L e_L}{\theta_H}\right) - \theta_H e_H + \psi(e_H) \\ \Leftrightarrow t_H &= -\theta_L e_L + t_L + \psi\left(\frac{\theta_L e_L}{\theta_H}\right) + \theta_H e_H - \psi(e_H) \\ \Leftrightarrow t_H &= -\theta_L e_L - \frac{1-\beta}{\beta} t_H + \psi\left(\frac{\theta_L e_L}{\theta_H}\right) + \theta_H e_H - \psi(e_H) \\ \Leftrightarrow t_H &= \beta \left(-\theta_L e_L + \psi\left(\frac{\theta_L e_L}{\theta_H}\right) + \theta_H e_H - \psi(e_H) \right) \end{aligned}$$

Second to last line because $t_L = -\frac{1-\beta}{\beta} t_H$ from the binding BC

And $t_L = -\frac{1-\beta}{\beta} t_H$, so

$$\begin{aligned} t_L &= -\frac{1-\beta}{\beta} t_H = -\frac{1-\beta}{\beta} \beta \left(-\theta_L e_L + \psi\left(\frac{\theta_L e_L}{\theta_H}\right) + \theta_H e_H - \psi(e_H) \right) \\ &= -(1-\beta) \left(-\theta_L e_L + \psi\left(\frac{\theta_L e_L}{\theta_H}\right) + \theta_H e_H - \psi(e_H) \right) \end{aligned}$$

Using the rewritten with the new expressions for t_L, t_H , we can omit the ICH:

$$\begin{aligned} &\beta u \left(\theta_L e_L + (1-\beta) \left(-\theta_L e_L + \psi\left(\frac{\theta_L e_L}{\theta_H}\right) + \theta_H e_H - \psi(e_H) \right) - \psi(e_L) \right) \\ &+ (1-\beta) u \left(\theta_H e_H - \beta \left(-\theta_L e_L + \psi\left(\frac{\theta_L e_L}{\theta_H}\right) + \theta_H e_H - \psi(e_H) \right) - \psi(e_L) \right) \end{aligned}$$

Which is equivalent to:

$$\begin{aligned} &\beta u \left(-\beta \theta_L e_L + (1-\beta) \left(\psi\left(\frac{\theta_L e_L}{\theta_H}\right) + \theta_H e_H - \psi(e_H) \right) - \psi(e_L) \right) \\ &+ (1-\beta) u \left((1-\beta) \theta_H e_H - \beta \left(-\theta_L e_L + \psi\left(\frac{\theta_L e_L}{\theta_H}\right) - \psi(e_H) \right) - \psi(e_L) \right) \end{aligned}$$

Since

$$\theta_L e_L + (1 - \beta)(-\theta_L e_L) = -\beta \theta_L e_L$$

And in the second part of the max:

$$\theta_H e_H - \beta \theta_H e_H = (1 - \beta) \theta_H e_H$$

For simplicity, define the inner part of the utilities, so to say, as:

$$\Gamma \equiv -\beta \theta_L e_L + (1 - \beta) \left(\psi \left(\frac{\theta_L e_L}{\theta_H} \right) + \theta_H e_H - \psi(e_H) \right) - \psi(e_L)$$

$$\Delta \equiv (1 - \beta) \theta_H e_H - \beta \left(-\theta_L e_L + \psi \left(\frac{\theta_L e_L}{\theta_H} \right) - \psi(e_H) \right) - \psi(e_L)$$

Problem is then

$$\max \beta u(\Gamma) + (1 - \beta) u(\Delta)$$

FOC for e_H is

$$e_H: \beta(1 - \beta)(\theta_H - \psi'(e_H))u'(\Gamma) + (1 - \beta)(1 - \beta)\beta(\theta_H - \psi'(e_H))u'(\Delta) = 0$$

$$\Leftrightarrow (\theta_H - \psi'(e_H))u'(\Gamma) + (1 - \beta)(\theta_H - \psi'(e_H))u'(\Delta) = 0$$

$$\Leftrightarrow \theta_H - \psi'(e_H) + (1 - \beta)(\theta_H - \psi'(e_H)) = 0$$

$$\Leftrightarrow \theta_H - \psi'(e_H) = 0$$

Since $u'(\Gamma), u'(\Delta) > 0$

Implying that

$$\theta_H = \psi'(e_H)$$

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$$\max \beta u(\theta_L e_L - t_L - \psi(e_L)) + (1 - \beta) u(\theta_H e_H - t_H - \psi(e_H))$$

$$\beta t_L + (1 - \beta) t_H \geq 0 \quad (BC)$$

$$\theta_L e_L - t_L - \psi(e_L) \geq \theta_L e_H - t_H - \psi(e_H) \quad (ICL)$$

$$\theta_H e_H - t_H - \psi(e_H) \geq \theta_H e_L - t_L - \psi(e_L) \quad (ICH)$$

Now, we just have a regular SB problem, i.e., one where the low (high) type must not choose the other type's contract by providing high (low) effort.

Problem 2

Consider the following adverse selection problem: There are many identical firms that can hire workers. Each firm produces the same output using an identical constant returns to scale technology in which labor is the only input. The firms are risk neutral, seek to maximize their expected profits, and act as price takers. The price of the firms' output is 1.

Workers differ in the number of units of output they can produce if hired by a firm. A worker of type θ can produce θ units of output, where θ is distributed uniformly on $[0, 1]$. The total measure of workers is 1. Workers' productivity levels are unobservable by the firms.

A worker can choose to work either at a firm or at home. A worker of type θ can earn $r(\theta) = \frac{4}{5}\theta^2$ in home production. A worker seeks to maximize his earnings from his labour by choosing where to work. Assume that if he is indifferent, a worker chooses to work in a firm.

1. Find the Pareto Optimal allocation of workers.
2. Find the competitive equilibrium.

Suppose instead that there are only two types of workers, $\theta_H = 1$ and $\theta_L = \frac{1}{2}$ and that $r(\theta_H) = r(\theta_L) = 0$. Workers' productivity levels are still unobservable by the firms. The fraction of workers who are type θ_H is λ .

Suppose jobs differ in the task level t required of the worker, and that the task does not affect the worker's productivity. The utility of a type θ worker who receives wage w and faces task level $t \geq 0$ is $u(w, t|\theta) = w - \frac{t^2}{\theta}$.

The players engage in the following two-stage game (which is a screening game):

- Stage 1: Two firms simultaneously announce sets of offered wage-task pairs (w, t) . Each firm may announce any finite number of such pairs.
- Stage 2: Given the offers made by the firms, workers of each type choose whether to accept a wage-task pair and, if so, which one. Assume that if a worker is indifferent between two wage-task pairs, he chooses the one with the lower task level and that he accepts employment if he is indifferent about doing so. If a worker's most preferred wage-task pair is offered by both firms, each is accepted with probability $\frac{1}{2}$.
3. What will be the wage paid to type θ_L , respectively type θ_H , in a separating SPNE?
4. Which task level will type θ_L , respectively type θ_H , engage in in a separating SPNE?

1

$$r(\theta) \leq \theta \Leftrightarrow \frac{4}{5}\theta^2 \leq \theta \Leftrightarrow 0 \leq \theta \leq \frac{5}{4}$$

And because $\theta \in [0, 1]$, the Pareto Optimal allocation of workers is therefore that every worker should work

2

The workers who are willing to work are the ones paid more than their productivity:

$$r(\theta) \leq w \Leftrightarrow \frac{4}{5}\theta^2 \leq w \Leftrightarrow \theta \leq \sqrt{\frac{5}{4}w}$$

Expected productivity of these workers:

$$w^* = \mathbb{E}[\theta | r(\theta) \leq w] = \mathbb{E}\left[\theta \left| \frac{4}{5}\theta^2 \leq w \right.\right] = \mathbb{E}\left[\theta \leq \sqrt{\frac{5}{4}w}\right] = \frac{1}{2}\sqrt{\frac{5}{4}w}$$

Because of uniform distribution

This implies that

$$w = \frac{1}{2}\sqrt{\frac{5}{4}w} \Leftrightarrow w \in \left\{0, \frac{5}{16}\right\}$$

Set of workers

$$\Theta = \{\theta | r(\theta) \leq w\} = \left\{\theta \left| \frac{4}{5}\theta^2 \leq w \right.\right\} = \left\{\theta \left| \frac{4}{5}\theta^2 \leq \frac{5}{16} \right.\right\} = \left\{\theta \left| \theta \leq \frac{5}{8} \right.\right\} = \left[0, \frac{5}{8}\right]$$

Final step:

$$w^* = \mathbb{E}[\theta | \theta \in \Theta] = \mathbb{E}\left[\theta \left| \theta \in \left[0, \frac{5}{8}\right] \right.\right] = \frac{1}{2} \cdot \frac{5}{8} = \frac{5}{16}$$

So, from this calculation, $w^* = \frac{5}{16}$ and therefore, this is a competitive equilibrium

It Pareto dominates the other equilibrium, $w = 0$, since more workers work, and they receive a positive wage

3

Wage must be equal to productivity since profit is $\theta_i - w_i = 0, i \in \{L, H\}$ and thus, $\theta_i = w_i$

For low type workers $\theta_L = w_L = \frac{1}{2}$

For high type $\theta_H = w_H = 1$

Profits are zero since if they were positive, one firm could gain all market share by offering ε higher wage

From slideshow L7:

We study two types of contracts (pooling and separating) and look for SPNE. Solving for SPNE will consist of a series of sub-results:

- ① In any equilibrium (pooling or separating) both firms must earn zero profits.
- ② No pooling equilibria exist. Therefore any equilibrium must be a separating equilibrium. So we focus on these.
- ③ If (w_L, t_L) and (w_H, t_H) are the contracts signed by the low- and high-ability workers in a separating equilibrium, then $w_L = \theta_L$ and $w_H = \theta_H$, i.e. both contracts yield zero profits.
- ④ In any separating equilibrium the low-ability workers accept contract $(\theta_L, 0)$; i.e. they receive the same contract as when no informational asymmetries are present.
- ⑤ In any separating equilibrium the high-ability workers accept contract $(\theta_H, \hat{t}_H)$, where $\hat{t}_H$ satisfies $\theta_H - c(\hat{t}_H, \theta_L) = \theta_L - c(0, \theta_L)$.

4

We use result 5 where the task level of the low type is $t_L = 0$:

$$w_L - \frac{t_L^2}{\theta_L} = w_H - \frac{t_H^2}{\theta_L} \Leftrightarrow \frac{1}{2} - \frac{0}{\frac{1}{2}} = 1 - \frac{t_H^2}{\frac{1}{2}} \Leftrightarrow \frac{1}{2} = 1 - 2t_H^2 \Leftrightarrow t_H = \frac{1}{2}$$

The reason why we have the ICL binding in Screening and not the ICH as we have seen in Adverse Selection is that the task level of the low type is set exactly so high that the low type is indifferent between his own contract and pretending to be the high type. If the ICL did not bind, the firm could lower the low type's task level until it did bind.

And just to be clear, the difference between adverse selection and screening is that in adverse selection, we study the choices taken by one party—an agent—which has more information than the other party—principal—typically about his abilities: type good or bad. In screening, we study the actions taken by the principal in order to distinguish between different types of agents.

In any equilibrium of the screening game, low-ability workers accept contract $(\theta_L, 0)$ and high ability workers $(\theta_H, \hat{t}_H)$, where $\hat{t}_H$ satisfies $\theta_H - c(\hat{t}_H, \theta_L) = \theta_L - c(0, \theta_L)$.

Hence, to prevent low ability workers from pretending to be high ability workers, the principal makes high ability workers perform an “unproductive” task.